PROFESSIONAL STATISTICIANS SOCIETY OF NIGERIA

THE BOOK OF ABSTRACT

9TH INTERNATIONAL CONFERENCE

THEME: ADVANCING STATISTICAL LITERACY IN AI APPLICATIONS: NIGERIA'S STRATEGY FOR DIGITAL TRANSFORMATION

VENUE: FEDERAL POLYTECHNIC, ADO-EKITI, EKITI STATE, NIGERIA

AUGUST, 2025

COMPARATIVE EVALUATION OF MACHINE LEARNING ALGORITHMS FOR THREE-CLASS DIABETES CLASSIFICATION USING NHANES DATA

T.M. Ayeni; E.A. Odukaya; A.O. Ilesanmi
Northeastern University, Toronto, Canada
Ekiti State University, Ado-Ekiti
Corresponding Author: Taiwo Ayeni

Background: Early detection of diabetes and prediabetes is critical for preventing long-term complications, including cardiovascular disease, neuropathy, and nephropathy. Type 2 diabetes affects over 537 million adults globally, with nearly half remaining undiagnosed. Machine learning offers potential for improving classification accuracy, but comprehensive comparative evaluations across multiple algorithms for three-class diabetes status prediction remain limited, particularly studies focusing on minimal feature sets suitable for resource-constrained settings. Objective: This study evaluates and compares four machine learning algorithms, Multinomial Logistic Regression (MLR), Decision Trees, Random Forest, and XGBoost, for classifying individuals into normal, prediabetic, and diabetic categories using readily available clinical indicators that can be obtained in primary care settings. Methods: Using data from the National Health and Nutrition Examination Survey (NHANES) with 2,029 participants (62.0% normal, 26.7% prediabetic, 11.3% diabetic), we employed Recursive Feature Elimination with 10-fold cross-validation to identify six key predictors from 27 candidate variables: fasting glucose, age, diabetes history, insulin level, waist circumference, and systolic blood pressure. Models were trained on a 70/30 stratified split and evaluated using overall accuracy, Cohen's Kappa, class-specific sensitivity, specificity, positive predictive value, and balanced accuracy. Results: Random Forest achieved the highest performance with 78.1% accuracy and Cohen's Kappa of 0.471, significantly outperforming XGBoost (72.6%), MLR (73.0%), and Decision Trees (72.4%). All models demonstrated excellent specificity for diabetes detection (>97%), minimizing false positives. Normal class identification showed robust sensitivity (87-88%) across algorithms. However, all models showed consistent challenges in prediabetes classification, with sensitivity ranging from 38-42%, reflecting the transitional nature of this metabolic state where physiological markers overlap with both normal and diabetic populations. Conclusions: Our comparative evaluation validates Random Forest as the optimal algorithm for three-class diabetes classification using NHANES data. The high performance achieved with only six routine clinical features obtainable with basic equipment in primary healthcare facilities suggests potential applicability in diverse clinical settings, including those with limited laboratory infrastructure. Future work should validate these findings in diverse populations, explore techniques to improve prediabetes detection, and develop implementation strategies for real-world deployment.

Keywords: Diabetes classification; Random Forest; NHANES; Machine Learning Comparison; Prediabetes detection; Multi-lass Classification